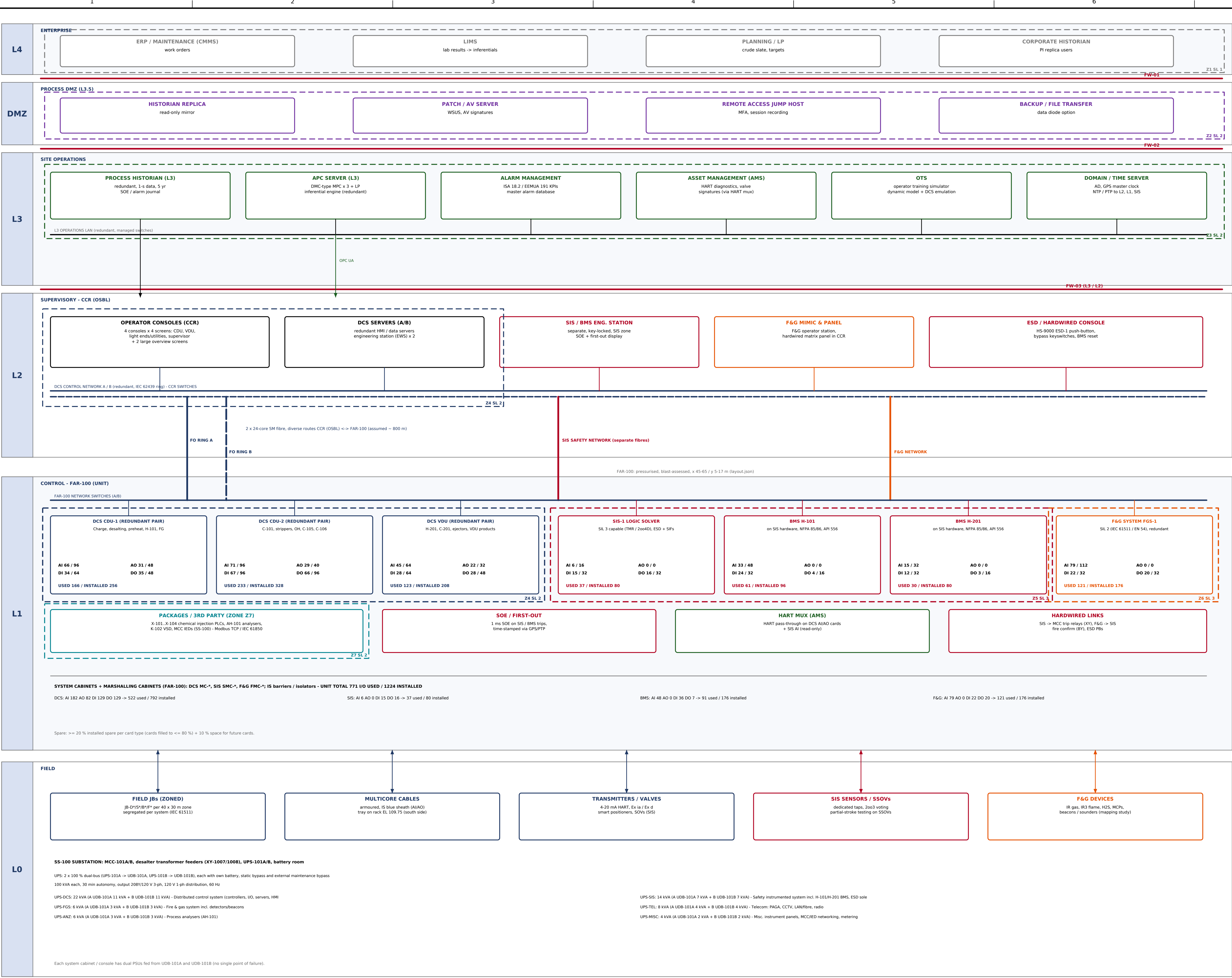




IEC 62443 ZONES

Z1 Enterprise (L4) - SL-T 1

Z2 Process DMZ (L3.5) - SL-T 2

Z3 Site operations (L3) - SL-T 2

Z4 BPCS / DCS (L2-L1) - SL-T 2

Z5 SIS / BMS (L2-L1) - SL-T 3

Z6 F&G (L2-L1) - SL-T 3

Z7 Packages / MCC / analysers - SL-T 2

CONDUITS

C1 Z1-Z2: FW-01 (L4/DMZ)

Historian replica read-only (PI-to-PI), patch/AV distribution, no inbound to L3

C2 Z2-Z3: FW-02 (DMZ/L3)

Historian push L3->DMZ only; remote access via jump host + MFA, session recorded

C3 Z3-Z4: FW-03 (L3/L2)

OPC UA (APC SP write, historian read), AMS HART pass-through, NTP; whitelisted

C4 Z4-Z5: SIS gateway (L2/SIS)

Read-only SIS -> DCS status/first-out via SIL-rated gateway; no DCS writes (bypass/reset by hardwired keyswitch)

C5 Z4-Z6: F&G gateway

Read-only F&G -> DCS mimic; F&G executive actions hardwired / SIS

C6 Z4-Z7: Package gateway

Modbus TCP / IEC 61850 (MCC IEDs, packages X-101..104, AH-101) via industrial firewall

ARCHITECTURE NOTES

1.

DCS: 3 redundant controller pairs (CDU-1, CDU-2, VDU) with redundant I/O comms and power; controller load <= 60 % CPU at FEED.

2.

SIS: one SIL 3-capable logic solver (TMR or 2oo4D), physically and logically separate from the DCS (IEC 61511 cl. 11.2); BMS H-101/H-201 configured as separate applications on SIS hardware.

3.

F&G: dedicated SIL 2 system; confirmed fire / gas executive actions per C&E CFU-000-IC-CE-001.

4.

Time synchronisation: GPS master clock (L3) - NTP / PTP to servers, controllers, SIS SOE (1 ms).

5.

Cyber: IEC 62443-3-2 risk assessment at detailed design; zones / conduits as shown; no direct L4 to L2 traffic; USB lock-down; whitelisting on EWS.

6.

I/O counts from data/i\_o\_list.json (CFU 000-IC-IOL-001).

| REV                                                     | DATE       | DESCRIPTION              |            |
|---------------------------------------------------------|------------|--------------------------|------------|
| A                                                       | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST                           |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT              |            |                          |            |
| ICS ARCHITECTURE BLOCK DIAGRAM                          |            |                          |            |
| DCS / SIS / BMS / F&G - PURDUE LEVELS & IEC 62443 ZONES |            |                          |            |
| DISCIPLINE                                              | STAGE      | SCALE                    | SHEET      |
| INSTRUMENTATION                                         | FEED       | NTS                      | 1 OF 1     |
| DRAWN                                                   | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                             | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                          |            |                          | REV        |
| CFU-000-IC-BLK-001                                      |            |                          | A          |